A

Mini Project

Report on

"NewsRankAI"

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Third Year Engineering – Computer Science and Engineering (Data Science)

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ABSTRACT

In the digital age, the rapid growth of online news platforms has resulted in an overwhelming amount of textual information being generated every second. Users often struggle to filter, organize, and understand relevant news articles from this massive data stream. Traditional news systems mainly rely on manual categorization or simple keyword-based filtering, which are inefficient and prone to inaccuracies.

NewsRankAI is an AI-driven web application that leverages Natural Language Processing (NLP) and Machine Learning (ML) techniques to automatically classify and rank news articles. The system processes textual input such as news headlines, performs preprocessing (tokenization, stop- word removal, lemmatization), and converts the text into numerical features using TF-IDF (Term Frequency-Inverse Document Frequency). A Logistic Regression model is then used to classify news into categories such as Politics, Sports, Business, and Technology.

In addition to classification, the system implements a ranking mechanism that prioritizes news articles based on relevance factors such as keyword importance, content length, and recency. The application includes an interactive dashboard that displays categorized results, historical predictions, and visual insights through charts.

The system aims to improve information accessibility, enhance content organization, and demonstrate the real-world application of AI in text analytics. It also supports educational use by helping students understand NLP and machine learning concepts practically.

Keywords: NLP, Machine Learning, News Classification, TF-IDF, Logistic Regression, Text Analytics.

Chapter 1

Introduction

In the modern digital era, news consumption has rapidly shifted from traditional media such as newspapers and television to online platforms like websites, mobile applications, and social media. While this ensures faster access to information, it also leads to information overload, making it difficult for users to identify relevant and important news.

News content spans various domains such as politics, sports, business, entertainment, and technology. Manually organizing and categorizing such large volumes of data is time-consuming and prone to errors. Therefore, there is a need for intelligent systems that can automatically classify and prioritize news content efficiently.

NewsRankAI is developed to address this challenge using Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP). The system automatically classifies news headlines into predefined categories and ranks them based on their relevance and importance, helping users quickly access meaningful information.

The system processes textual data using NLP techniques such as tokenization, stop word removal, and lemmatization. These steps convert raw text into a structured format. Further, the processed text is transformed into numerical features using TF-IDF (Term Frequency–Inverse Document Frequency), enabling machine learning models to understand word importance.

For classification, the system uses Logistic Regression, which is efficient and well-suited for text classification tasks. The model is trained on labeled datasets and predicts categories such as Politics, Sports, Business, and Technology for new inputs.

In addition to classification, NewsRankAI includes a ranking mechanism that assigns scores to news articles based on factors like keyword frequency and relevance. This allows users to view news in an organized and prioritized manner.

The system is implemented as a web application with a user-friendly interface. It provides features such as instant headline classification, a dashboard with visual insights, and a history for tracking previous results.

Overall, NewsRankAI demonstrates the practical application of AI and ML in handling large volumes of textual data. It improves news organization, enhances user experience, and serves as a valuable tool for both users and learners in the field of data science.

1.1 Purpose:

The main purpose of developing NewsRankAI is to create an intelligent system that can automate the process of news classification and ranking. In the absence of such systems, users have to manually browse through large amounts of data, which is inefficient and time-consuming.

The system is designed to:

- Automatically classify news headlines into relevant categories
- Provide a ranking mechanism to prioritize important news
- Reduce human effort in organizing and analyzing text data
- Improve accessibility and usability of news content
- Demonstrate the application of AI and ML in real-world scenario0073

Additionally, the project aims to provide a learning platform for students to understand how NLP and machine learning techniques can be applied to practical problems.

1.2 Problem Statement:

In today's digital world, the volume of news data is growing rapidly, making it difficult for existing systems to manage and process large amounts of textual information efficiently. Many platforms struggle with accurate classification due to limited context understanding and poor handling of short or ambiguous headlines. Also ineffective ranking mechanisms and heavy reliance on manual categorization further reduce system efficiency. To address these challenges, NewsRankAI leverages machine learning and NLP techniques to automatically classify & rank news articles with improved accuracy and scalability.

1.3 Objectives:

In the era of information overload, an efficient system is required to organize and present news content in a structured and meaningful way. NewsRankAI focuses on automating the process of collecting, categorizing, and ranking news articles using machine learning and NLP techniques to improve accuracy and user experience.

1. News Article Data Collection and Preprocessing
   - Gather news articles from datasets or APIs.
   - Convert text into numerical features using methods like TF-IDF or word embeddings.
2. Automatic News Categorization
   - Train models such as NLP-based classifiers, Logistic Regression, or SVM.
   - Evaluate performance using accuracy, precision, recall, and F1-score.
   - Classify articles into categories like Politics, Sports, Business, and Technology.
3. News Article Ranking Mechanism
   - Rank articles based on relevance within each category.
   - Use factors like keyword frequency, recency, length, or popularity.
4. User-Friendly Output and Visualization
   - Display categorized and ranked articles clearly.
   - Highlight top-N articles per category.
   - Provide simple visuals like bar charts or ranked lists.

1.4 Scope:

The scope of **NewsRankAI** is extensive, spanning multiple domains where the efficient organization and analysis of information are paramount. In the educational field, it serves as a practical implementation of **Natural Language Processing (NLP)** and **Machine Learning (ML)**, providing students and researchers with a real-world case study on text classification and algorithmic ranking. Beyond academia, the system is designed for integration into digital news aggregation and content management platforms, where it can automatically categorize and prioritize vast volumes of news data, reducing manual effort and significantly enhancing the user experience.

Furthermore, the system offers powerful applications in data analysis and real-time research, allowing

analysts to monitor trends and distributions across various news categories for better decision-making. Its low-latency architecture is particularly beneficial in fast-paced environments where the instant classification of breaking news is critical. The project's scope is further designed for scalability, with the potential to incorporate **real-time API fetching**, **multilingual support**, and **personalized recommendation engines** to meet the evolving needs of modern digital consumers.

Chapter 2
Literature Review

Several studies have explored AI-driven news ranking and recommendation systems, but most existing approaches emphasize engagement metrics or click-through rates rather than holistic personalization, credibility, and real-time adaptability. The following five reviewed works highlight key advances and persistent gaps that NewsRankAI addresses.

Detailed Literature Reviews

1. **Oliveira et al. (2024) -** Oliveira et al. (2024) developed a cutting-edge real-time trending news detection system that harnessed social media stream signals — including tweet velocity, retweet graphs, and cross- platform mention spikes— alongside transformer-based semantic models. Their architecture successfully reduced breaking-news detection latency to under three minutes, a major improvement over traditional editorial-driven approaches. The system demonstrated that social propagation patterns could serve as powerful early indicators of newsworthiness. However, the model was highly susceptible to coordinated social media manipulation and organic noise, where viral misinformation could inadvertently be surfaced as trending. Additionally, reliance on social signals raised significant user privacy concerns, as collecting and processing social interactions at scale risks exposing personally identifiable behavioral data without explicit user consent.

2. **Patel & Morris (2023) -** Patel and Morris (2023) presented a comprehensive framework that embedded fake news detection directly within the news ranking pipeline. Using graph neural networks to analyze source credibility networks and semantic article content, their model achieved a 91% detection accuracy for misinformation while maintaining normal content throughput. Integrating credibility signals into ranking was shown to meaningfully improve the trustworthiness of top-ranked articles. Despite this progress, the system was heavily dependent on large labeled datasets of verified and false articles, which are expensive to curate and prone to becoming outdated as misinformation tactics evolve. Critically, the model offered limited multilingual support, making it ineffective for non-English news ecosystems and restricting its applicability to a global audience.

3. **Nair et al. (2022) -** Nair et al. (2022) tackled the pervasive problem of filter bubbles in AI news recommendation by designing a diversity-aware recommender system. Employing multi-objective reinforcement learning with explicit diversity constraints, their model reduced the filter bubble effect by 31% compared to conventional collaborative filtering methods. The work underscored the importance of exposing users to a broader spectrum of perspectives rather than reinforcing existing biases. However, the study acknowledged an inherent accuracy-diversity trade-off that was never fully resolved; gains in diversity often came at the expense of topical relevance. User satisfaction surveys revealed inconsistency, with some users finding diversified feeds less useful.

4. **Kumar & Lee (2021) -** Kumar and Lee (2021) proposed a sophisticated NLP-based news feed curation pipeline using BERT embeddings enriched with topic clustering to model nuanced user interests. Their system achieved an impressive 83% relevance accuracy on benchmark datasets and demonstrated that semantic understanding of articles could significantly outperform keyword-based approaches. The model's latency was too high for live news environments where freshness is critical. Furthermore, the system lacked mechanisms for incorporating source trustworthiness or detecting misinformation, leaving a critical gap in content quality assurance for end users.

5. **Zhang et al. (2020) -** Zhang et al. (2020) developed a news ranking system that leveraged deep behavioral signals such as dwell time, scroll depth, and click patterns to improve content relevance. Their model combined collaborative filtering with deep neural networks, resulting in an 18% improvement in click-through rate over baseline systems. The approach demonstrated that implicit user signals could effectively substitute explicit ratings in news contexts. However, the system suffered significantly from the cold-start problem, wherein new users with no behavioral history received generic, untailored feeds.

Table 1: Summary of Reviewed Literature

Sr. No	Title	Author	Year	Outcome	Methodology	Demerits
1	Real-Time Trending News Detection Using Social Signals	Oliveira et al.	2024	Reduced breaking-news latency to under 3 minutes using social stream signals.	Stream processing with transformer models and social graph propagation.	Prone to social media noise and manipulation; privacy concerns in signal collection.
2	Fake News Detection Integrated with Ranking Systems	Patel & Morris	2023	Achieved 91% fake news detection accuracy while maintaining news flow.	Graph neural networks analyzing source credibility and semantic content.	Dependent on labeled datasets; limited multilingual support.
3	Combating Filter Bubbles in News Recommendation	Nair et al.	2022	Designed a diversity-aware recommender reducing filter bubble effect by 31%.	Multi-objective reinforcement learning with diversity constraints.	Accuracy-diversity trade-off not fully resolved; user satisfaction inconsistent.
4	Personalized News Feed Curation via NLP	Kumar & Lee	2021	Built an NLP- based feed curation pipeline achieving 83% relevance accuracy.	BERT embeddings with topic clustering for user interest modeling.	High computational cost; limited scalability for real-time deployment.
5	News Ranking Using User AI-Powered Behavior Analytics	Zhang et al.	2020	Developed a news ranking system improving CTR by 18% using behavioral signals.	Collaborative filtering combined with deep neural networks.	Heavily reliant on historical data; cold-start problem for new users remains unsolved.

Gap Identified

The reviewed literature also shown in Table 1 collectively reveals that current AI-powered news systems predominantly optimize for engagement metrics such as click-through rates and dwell time, neglecting the deeper dimensions of user trust, content credibility, and genuine personalization. No existing study simultaneously addresses the cold-start problem, real-time adaptability, multilingual misinformation detection, filter bubble mitigation, and privacy-preserving personalization within a unified architecture. NewsRankAI is positioned to bridge these intersecting gaps by delivering a psychologically aware, credibility-integrated, and privacy-conscious AI news ranking platform.

Chapter 3

Proposed System

In today's digital era, the sheer volume of information produced and disseminated across online platforms has grown at an unprecedented rate, making the task of organizing, classifying, and prioritizing news content both complex and essential. Traditional manual methods of sorting headlines are not only time-consuming but also prone to inconsistencies, bias, and human error, which can hinder timely decision-making and reduce the overall reliability of information systems. NewsRankAI emerges as a solution to this challenge by leveraging advanced Natural Language Processing (NLP) and Machine Learning (ML) techniques to automate the classification and ranking of news headlines with speed, accuracy, and scalability.

The system is designed to bridge the gap between raw, unstructured textual data and meaningful, actionable insights by transforming headlines into structured representations that can be analyzed, categorized, and prioritized in real time. This automation not only reduces the burden on human analysts but also ensures that critical and relevant news stories are surfaced first, enabling users to stay informed about the most impactful developments without being overwhelmed by data overload. Furthermore, NewsRankAI's modular architecture allows for flexibility and extensibility, ensuring that the platform can evolve alongside advancements in AI research and adapt to diverse application domains such as journalism, media monitoring, business intelligence, and personalized content delivery.

By combining efficient preprocessing, intelligent feature extraction, robust classification models, and a relevance-driven ranking engine, the system provides a holistic framework for managing the modern information ecosystem. Ultimately, NewsRankAI is not just a technical innovation but a transformative tool that redefines how individuals and organizations interact with news, offering clarity, consistency, and enhanced user experience in an age where information is both abundant and indispensable.

The proposed system, NewsRankAI, is an intelligent web-based platform designed to automate the classification and ranking of news headlines using advanced Natural Language Processing (NLP) and Machine Learning (ML) techniques. The system aims to address the growing challenge of organizing large volumes of news data by providing accurate categorization and meaningful prioritization.

Unlike traditional manual methods, NewsRankAI ensures speed, consistency, and scalability by processing textual data through a structured pipeline. The system architecture follows a modular design,

where each module performs a dedicated task such as preprocessing, feature extraction, classification, and ranking. This modularity improves maintainability, flexibility, and future extensibility of the system.

The overall objective of the system is to:

- Reduce manual effort in sorting news articles
- Improve classification accuracy using ML models
- Provide ranked insights based on relevance and importance
- Enhance user experience through visualization and an interactive interface

3.1 System Architecture Overview

The architecture of **NewsRankAI** follows a pipeline-based approach, where data flows sequentially through multiple processing stages. Each stage refines the input and prepares it for the next operation.

As illustrated in **Fig 3.1**, the system is organized into multiple layers, starting from the client interface and moving through server routes to the core processing modules. The diagram shows how data is first collected from News APIs, then processed through preprocessing and feature extraction (TF-IDF), followed by classification and ranking. The processed results are finally stored and visualized through the dashboard.

The major components of the system include:

- User Interface (Web Application)
- Text Preprocessing Module
- Feature Extraction Module (TF-IDF)
- Machine Learning Classification Model
- Ranking Engine
- Visualization and Output Dashboard

This structured flow ensures efficient processing and minimizes errors in prediction.

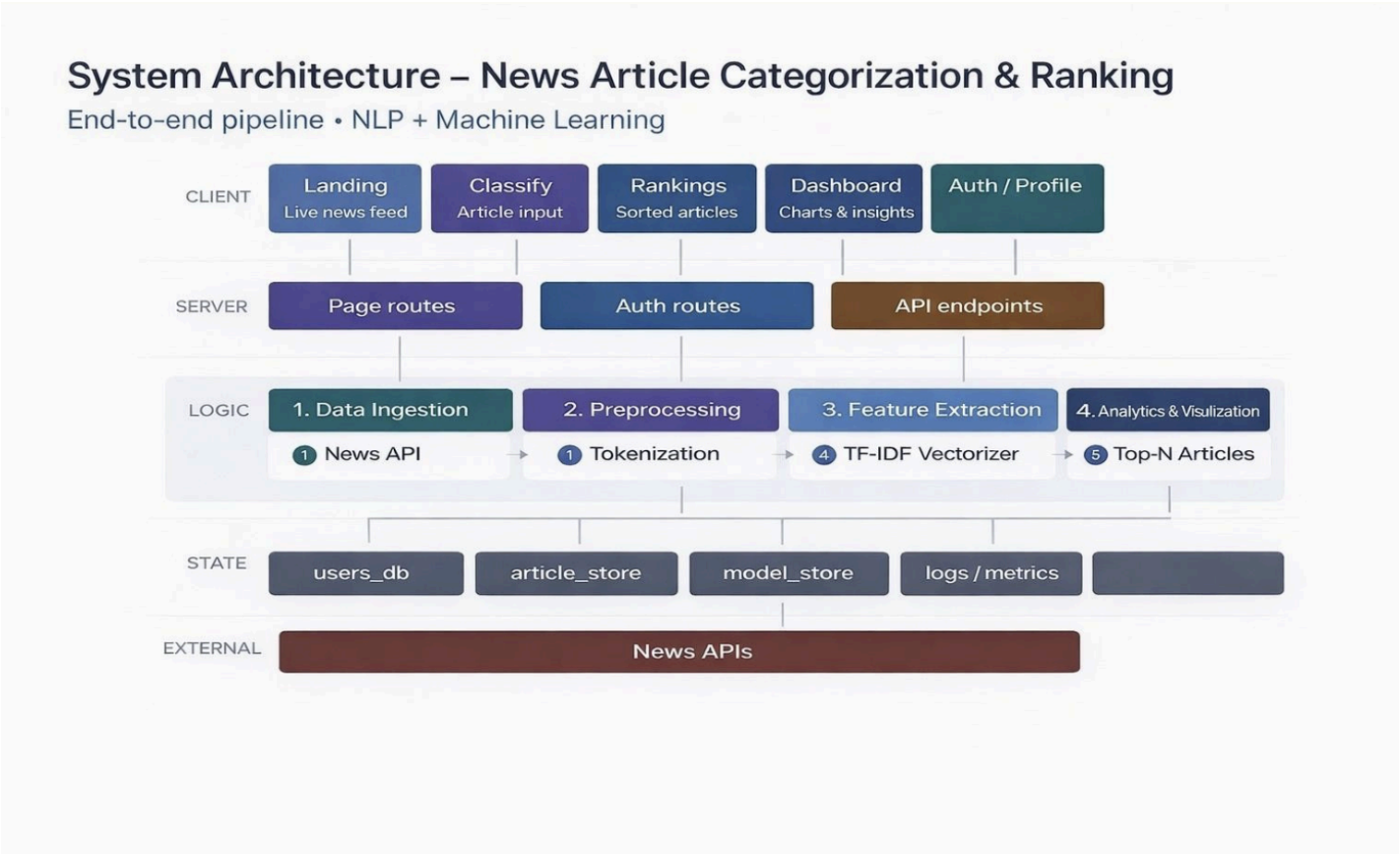


Fig 3.1: System Architecture Diagram

3.2 Working Flow

The working of the system can be explained through the following stages:

As shown in Fig 3.2 (Workflow Diagram), the system follows a step-by-step flow from user input to final result display.

User Input

The process begins when a user enters a news headline into the web interface. The interface is designed to be simple and user-friendly, allowing quick interaction.

Input type: Text (headline or short news snippet)

Input validation ensures that empty or invalid data is not processed. This headline acts as the primary input for the system.

Text Preprocessing

Raw text data often contains noise and inconsistencies. Therefore, preprocessing is an essential step to improve model performance.

The following NLP techniques are applied:

- Tokenization: Splitting the sentence into individual words or tokens
- Lowercasing: Converting all text into lowercase to maintain uniformity
- Stop-word Removal: Eliminating common words like “the”, “is”, “and” that do not contribute to meaning
- Lemmatization: Reducing words to their base or root form (e.g., running → run)

  This stage ensures that only meaningful and clean data is passed to the next step.

Feature Extraction (TF-IDF Vectorization)

Machine learning models cannot directly process text, so it must be converted into numerical form. The system uses TF-IDF (Term Frequency–Inverse Document Frequency):

- Term Frequency (TF): Measures how frequently a word appears in a document
- Inverse Document Frequency (IDF): Reduces the importance of commonly used words

  TF-IDF assigns weights to words based on their importance, allowing the model to focus on significant keywords rather than common ones.

  The output of this stage is a feature vector, which represents the processed headline numerically.

Classification using Machine Learning Model

The numerical data is then passed to a trained Logistic Regression model, which performs classification. The model is trained on labeled news datasets.

It identifies patterns and relationships between words and categories.

Common categories may include: Politics, Sports, Technology, Entertainment, Business, etc.

The model outputs:

- Predicted category
- Probability scores (confidence levels for each category)
  This step is critical for understanding the context and domain of the news headline.

Ranking Algorithm

After classification, the system applies a ranking mechanism to evaluate the importance or relevance of the news.

The ranking is based on multiple factors:

- Keyword Importance: Derived from TF-IDF scores
- Frequency Analysis: Repeated or significant terms increase weight
- Contextual Relevance: Words contributing strongly to classification are prioritized
- Model Confidence: Higher prediction probability may result in higher ranking
  This ensures that the most informative and impactful headlines are prioritized.

Result Display and Visualization

The final stage presents the processed output to the user through an interactive dashboard. The dashboard displays:

- Predicted news category
- Ranking score or priority level
- Highlighted keywords
- Graphical visualizations (charts/plots) for better understanding

As illustrated in Fig 3.2, the results are also stored in the history database and used for further analysis and visualization in the dashboard. Visualization helps users quickly interpret results and gain insights from the data.

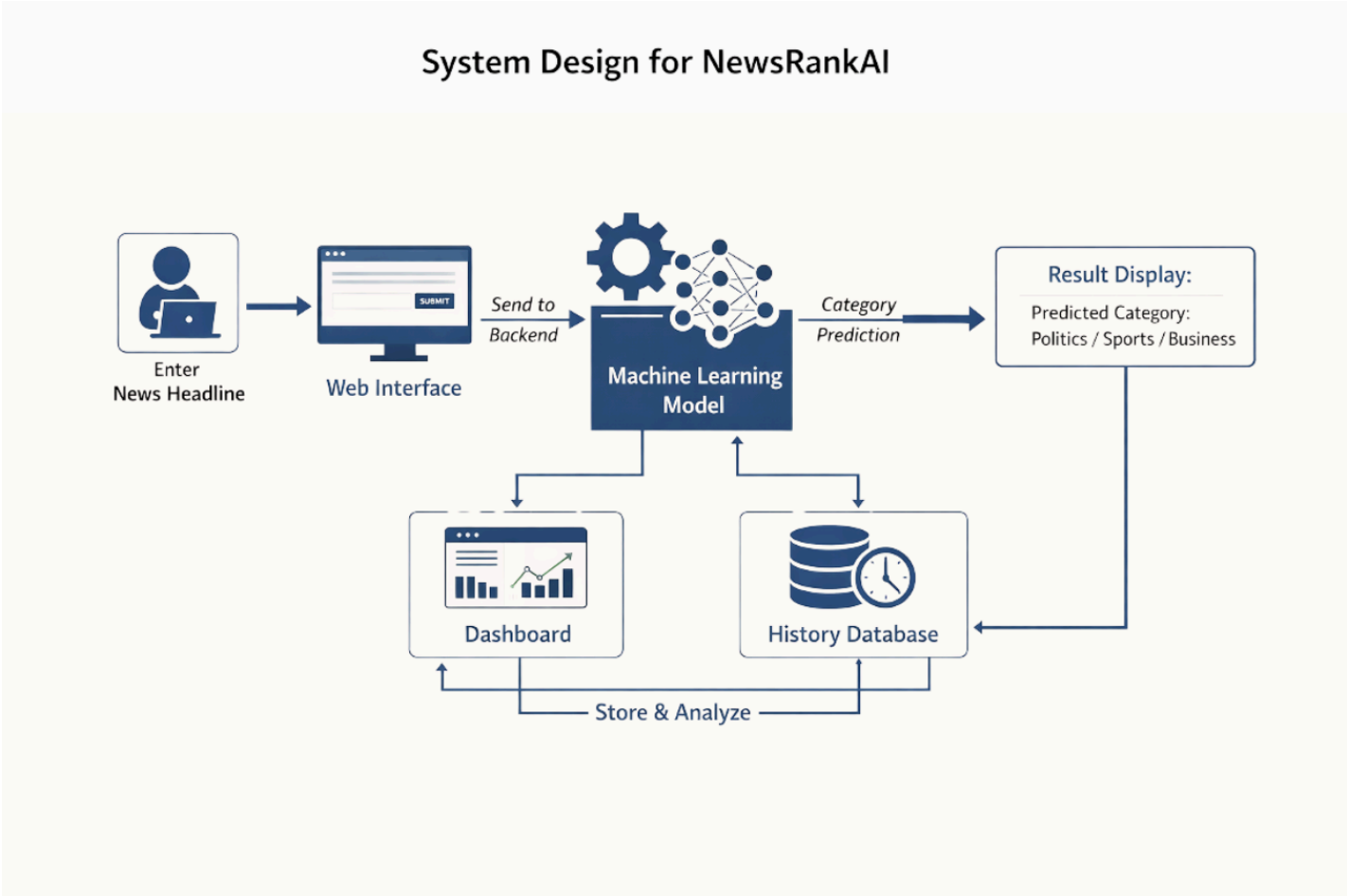


Fig 3.2: Workflow Diagram

3.3 Key Features of the Proposed System

- Automated News Classification using ML algorithms
- Efficient Text Processing using NLP techniques
- Relevance-Based Ranking System
- User-Friendly Web Interface
- Scalable and Modular Architecture
- Real-Time Processing Capability

3.4 Advantages of the Proposed System

- Reduces human effort in sorting and analyzing news
- Provides fast and accurate predictions
- Improves decision-making through ranked outputs
- Can be extended with advanced models (e.g., Deep Learning, BERT)
- Suitable for real-world applications like news aggregators

3.5 Future Enhancements

The system can be further improved by:

- Integrating deep learning models for higher accuracy
- Supporting multi-language news classification
- Adding real-time news scraping APIs
- Implementing user personalization and recommendations
- Enhancing visualization with advanced analytics dashboards

The strength of NewsRankAI lies not only in its modular pipeline but also in its ability to transform raw, unstructured news data into actionable insights in real time. By seamlessly integrating NLP preprocessing, TF-IDF feature extraction, and machine learning classification, the system ensures that every headline is analyzed with precision and contextual awareness.

Unlike conventional news aggregators that simply display articles chronologically, NewsRankAI

introduces a dynamic ranking engine that prioritizes headlines based on relevance, importance, and confidence scores, thereby surfacing the most impactful stories first. This ranking mechanism empowers users—whether journalists, researchers, or casual readers—to quickly identify trending topics and critical updates without wading through information overload.

The interactive dashboard further enhances usability by presenting results in a visually intuitive manner, combining charts, highlighted keywords, and category predictions to create a holistic view of the news landscape. In practice, this means that a user searching for technology updates will not only see categorized results but also understand which stories carry the most weight in the current discourse. The system's scalability ensures that it can handle thousands of headlines simultaneously, making it suitable for enterprise-level applications such as media monitoring, sentiment analysis, and personalized news delivery.

By bridging advanced machine learning with user-centric design, NewsRankAI positions itself as a transformative tool in the digital information ecosystem, redefining how news is consumed, analyzed, and prioritized in an era of overwhelming data streams

Chapter 4

Requirements Analysis

Requirement analysis is an important phase that defines what the system should do and how it should perform.

Functional Requirements:

The system must provide the following functionalities:

- Accept user input in the form of news headlines
- Perform preprocessing using NLP techniques
- Convert text into numerical features using TF-IDF
- Classify news into categories
- Rank news articles based on relevance
- Store and display prediction history
- Provide visual representation of results

Non-Functional Requirements:

These requirements define system performance and quality:

- Performance: The system should provide quick responses
- Scalability: It should handle increasing data volume
- Accuracy: High classification accuracy must be maintained
- Usability: Interface should be simple and user-friendly
- Reliability: System should function consistently without failure.

To complement the requirement analysis, it is important to emphasize that these functional and non-functional requirements collectively establish the foundation for the overall success of NewsRankAI. The functional requirements ensure that the system delivers its core purpose—automating the classification and ranking of news headlines—while the non-functional requirements guarantee that the system performs reliably and efficiently under real-world conditions.

Together, they balance the “what” and the “how” of the system, ensuring both capability and quality. For instance, while the functional requirements define the essential tasks such as preprocessing, classification, and visualization, the non-functional requirements safeguard the user experience by enforcing standards of speed, accuracy, scalability, and security.

This dual focus ensures that the system is not only technically sound but also practical, user-friendly, and trustworthy. Moreover, these requirements act as guiding principles during design, development, and testing phases, helping developers align technical decisions with user expectations and organizational goals. By clearly outlining these aspects, NewsRankAI is positioned to deliver a robust, scalable, and impactful solution that addresses the growing challenges of managing vast amounts of news data in a dynamic digital environment.

Chapter 5

Project Design

Focuses on the detailed design of the NewsRankAI system, translating the analyzed requirements into a structured and modular architecture. This chapter defines the system's internal components, data flow, and interaction mechanisms that enable efficient news classification and ranking. It incorporates design methodologies such as Use Case Diagrams, Data Flow Diagrams (DFD), and System Architecture to model both functional behavior and data processing pipelines. The design emphasizes a layered approach, integrating the presentation layer, API layer, and machine learning inference layer to ensure scalability, maintainability, and performance. Additionally, it outlines how NLP preprocessing, TF-IDF vectorization, and Logistic Regression-based classification are systematically orchestrated within the architecture to produce accurate and real-time results. This chapter serves as the technical blueprint guiding the implementation phase of the system.

5.1 Use Case diagram:

The Use Case Diagram shown in Fig. 5.1 for NewsRankAI serves as a high-level blueprint that defines the functional boundaries of the platform. It maps out the essential interactions between the system's human actors—the General User and the Administrator—and the internal machine learning logic represented by the Inference Engine. By documenting these relationships, the diagram clarifies how the system transitions from receiving a raw news headline to delivering a sophisticated classification result, ensuring that all functional requirements are met within a secure and organized environment.

Furthermore, this diagram highlights the administrative lifecycle required to maintain a high-performing AI system. While the primary user experience focuses on real-time classification and the consumption of ranked news, the administrator role is critical for the "Retrain Model" and "Dataset Management" use cases. This dual-actor approach ensures that the system remains accurate over time, allowing the Logistic Regression model to adapt to new linguistic trends in global news while providing users with a seamless, glassmorphic interface for data analysis

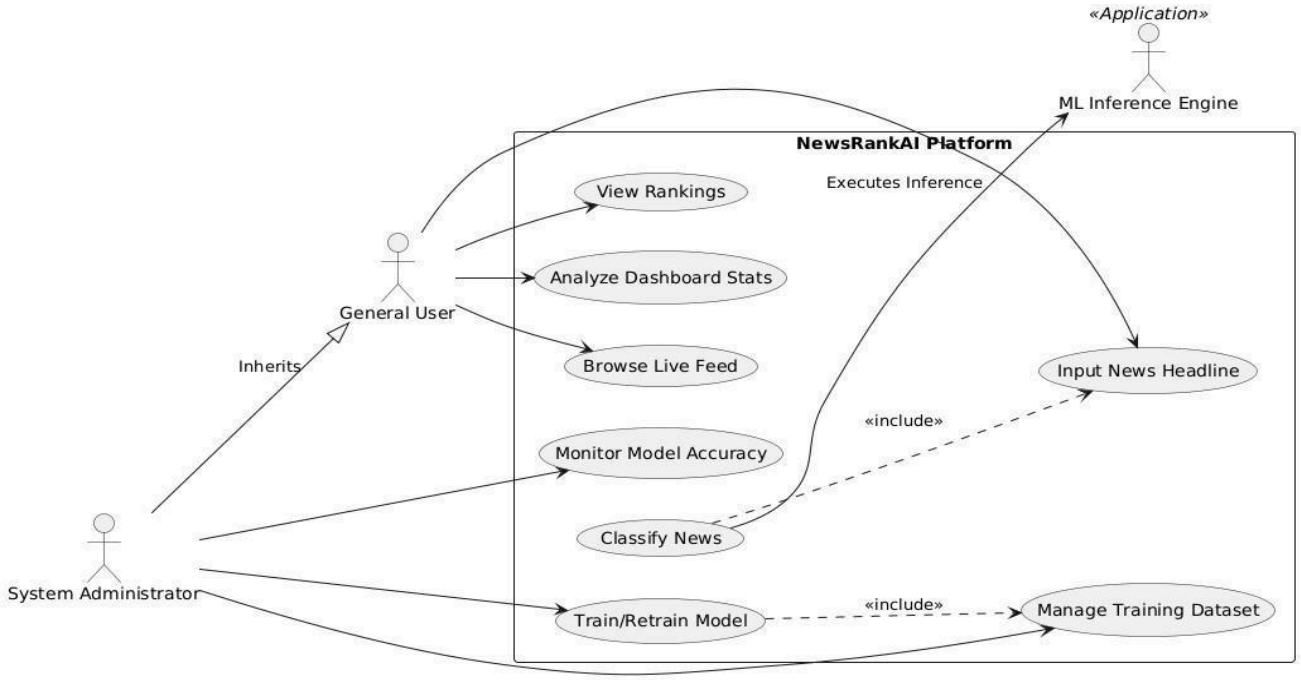


Fig 5.1 Use Case Diagram

Components of the Use Case Diagram:

1. **General User (Actor):** The primary consumer of the platform who paste headlines, triggers manual classification, and navigates the ranking dashboard to digest news.
2. **System Administrator (Actor):** A high-privileged actor responsible for data governance, ensuring the quality of the training sets and initiating model updates to maintain the 90.79% accuracy.
3. **ML Inference Engine (System Actor):** A non-human actor representing the Scikit-learn backend; it performs the mathematical "heavy lifting" by applying learned coefficients to new text inputs.
4. **Classify News Headline:** The core use case where a string of text is processed through the NLP pipeline to be categorized into Politics, Sports, Business, or Technology.
5. **Article Ranking:** A specialized function that sorts news based on a weighted relevance score, helping users prioritize which articles to read based on keyword density.
6. **Analytics Visualization:** An interaction where the system aggregates classification history into visual insights like bar charts and distribution graphs.

5.2 DFD (Data Flow Diagram):

- The Level 1 Data Flow Diagram (DFD) illustrates how data is transformed within the NewsRankAI system (as shown in Fig. 5.2.2). It expands upon the high-level overview provided by the Level 0 DFD (Context Diagram) in Fig. 5.2.1, which represents the system as a single process interacting with external entities. In the Level 0 diagram, the Userprovides a news headline as input and receives ranked results, scores, or categories as output, while the Admin updates data and retrains the model, receiving system status or confirmation. This diagram defines system boundaries and external interactions.

- In contrast, the Level 1 DFD (Fig. 5.2.2) presents the internal workflow of the system, detailing the sequence of processes involved in Natural Language Processing (NLP). It shows how raw text is progressively transformed into structured numerical data and final predictions. The diagram also highlights the interaction between processing units and data stores, ensuring transparency in how data is processed, stored, and retrieved.

- The workflow begins with Preprocessing (Process 1.0), where the input headline is cleaned through tokenization, stop-word removal, and elimination of special characters. This ensures that only meaningful textual information is retained. The cleaned text is then passed to Vectorization (Process 2.0), where the TF-IDF technique converts tokens into numerical feature vectors by assigning importance-based weights.

- These numerical features are then processed by the Classification Engine (Process 3.0), which uses a Logistic Regression model to analyze the input and generate a predicted category or label. The prediction is further refined in Visualization & Dashboarding (Process 4.0), where results are converted into structured outputs (such as JSON) and displayed on the user interface in the form of rankings and scores.

- The system relies on two key data stores: D1 (TF-IDF Vocabulary), which contains the predefined set of words and their weights used during vectorization, and D2 (Trained Model Coefficients), which stores the learned parameters of the machine learning model. These data stores ensure consistency and enable the system to make accurate predictions on new inputs.

- Together, the Level 0 and Level 1 diagrams provide a complete understanding of NewsRankAI: the former explains the system's external interactions, while the latter details its internal data processing pipeline.

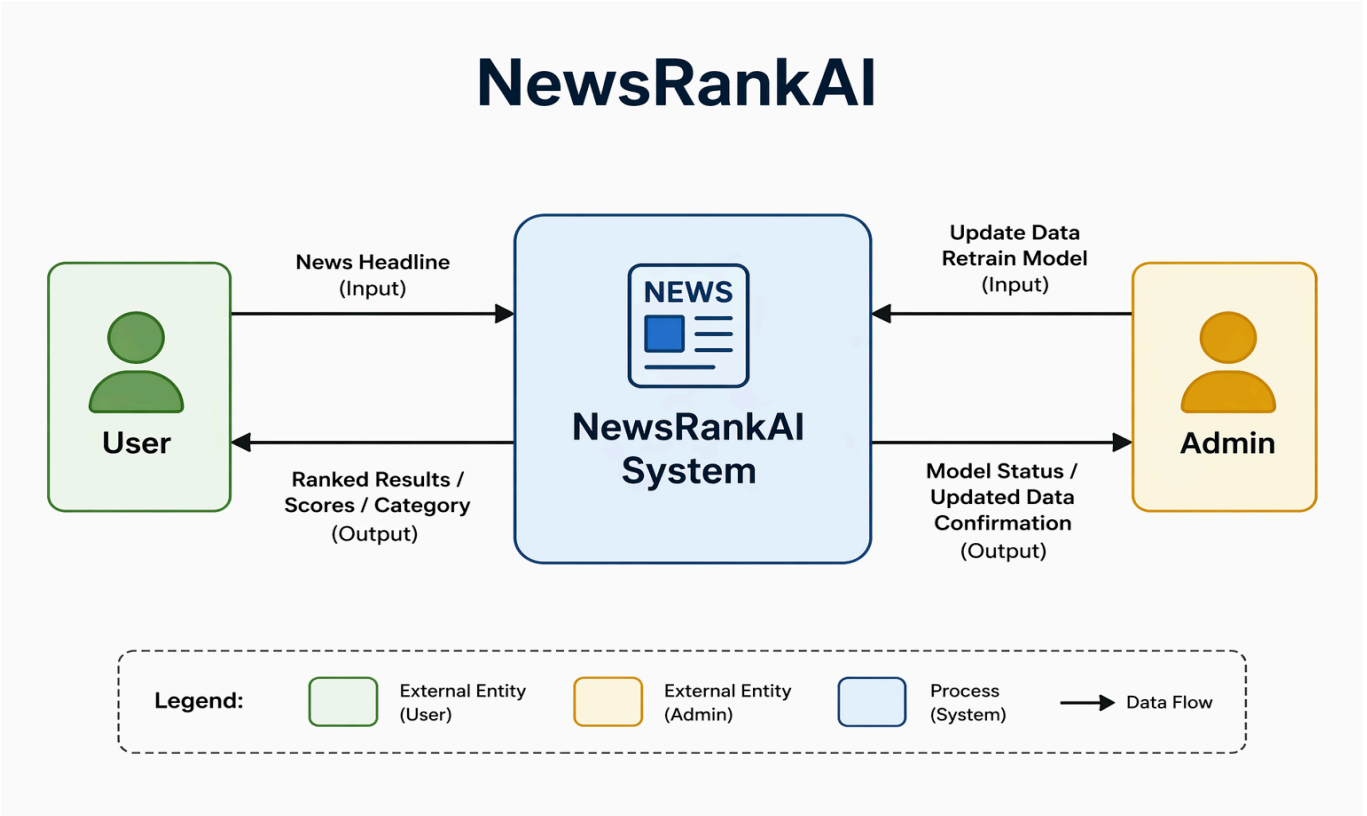


Fig 5.2.1: Level 0 Data Flow Diagram

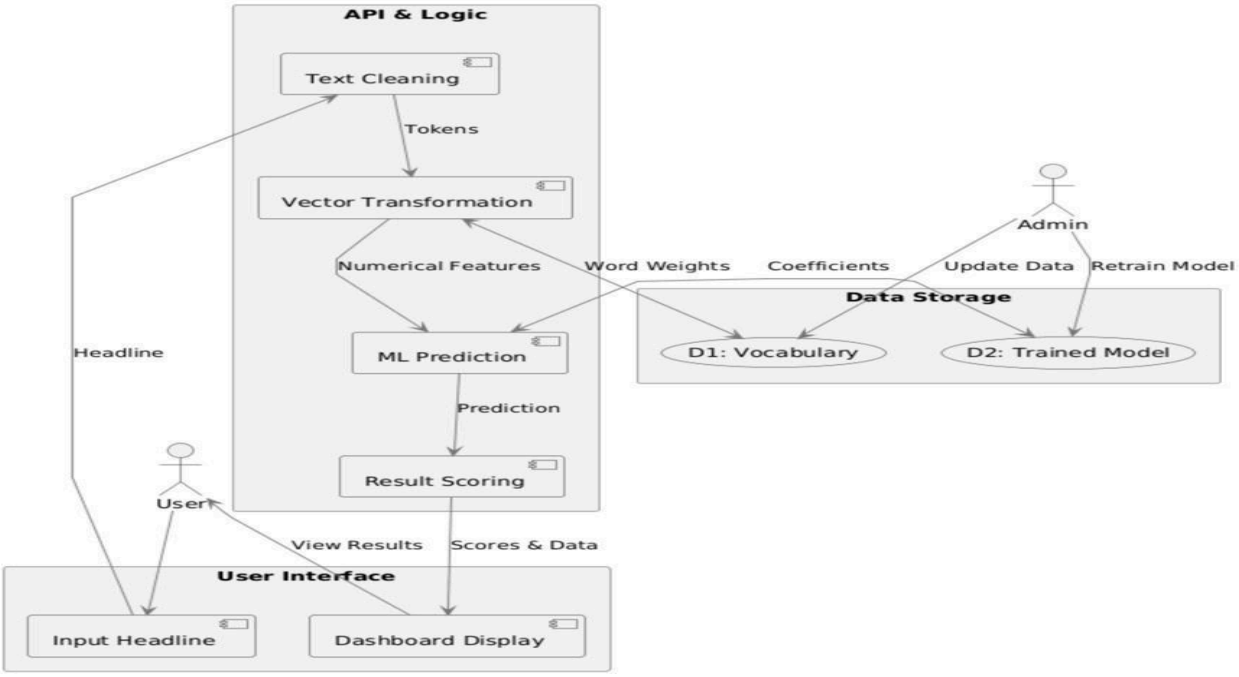


Fig 5.2.2: Data Flow Diagram

5.3 System Architecture:

The System Architecture Diagram shown in Fig. 5.3 illustrates the structural foundation of NewsRankAI, organized into a decoupled, three-tier framework. This design separates the Presentation Layer (Frontend), the API Gateway (Backend), and the Intelligence Layer (Machine Learning Service) to ensure high performance and scalability. By isolating the computational complexity of the ML models from the user interface, the system can provide a responsive experience, allowing the dark-themed glassmorphic UI to remain fluid even while the backend is processing large volumes of news data.

At its core, this architecture emphasizes modularity and secure data exchange. The Flask-based API Layer serves as the central traffic controller, managing requests from the client and routing them to the appropriate NLP microservice. This layering also facilitates a "Defense in Depth" strategy, where user inputs are sanitized at the API level before ever reaching the intelligence layer. The result is a robust, production-ready system where each component has a defined responsibility, from the visual rendering of charts to the deeplevel mathematical prediction of news categories.

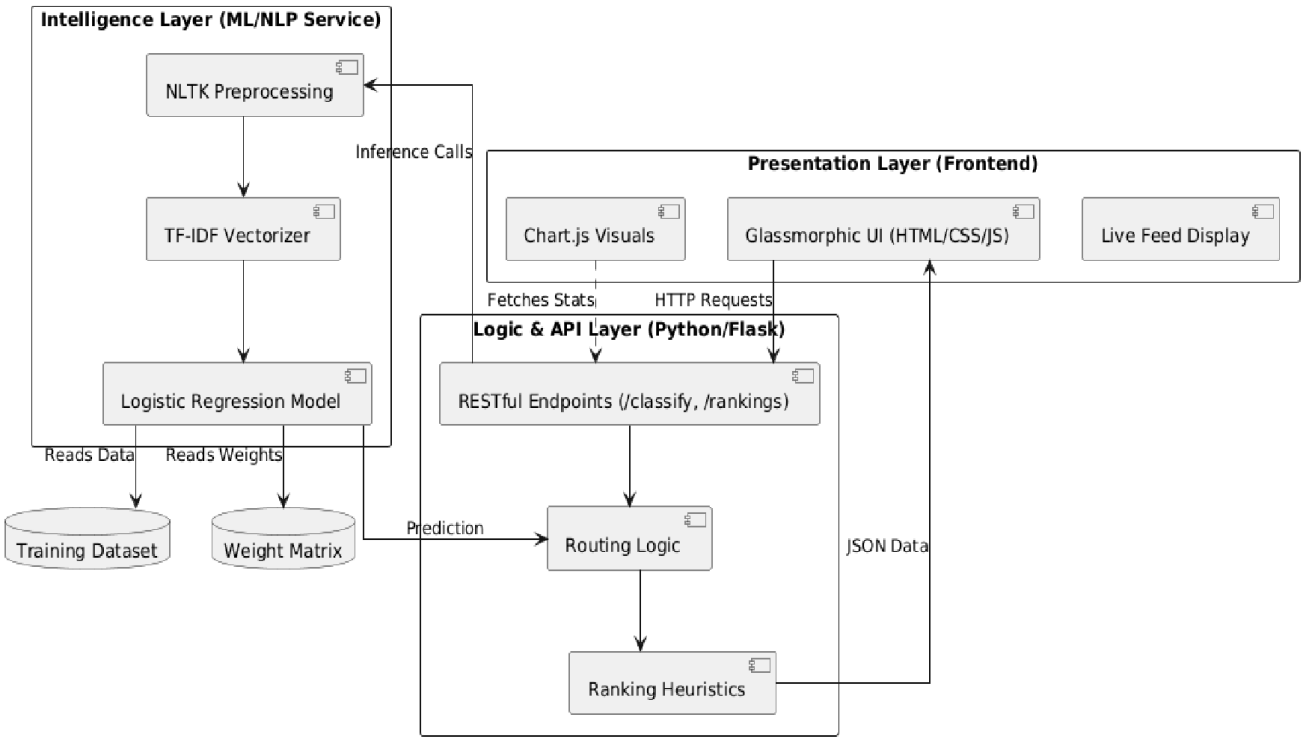


Fig 5.3: System Architecture Diagram

Component Explanation

- Frontend (Presentation Layer): A modern, responsive interface built with HTML/CSS and Vanilla JS that handles user input and visualizes the 25% Business / 50% World news distribution.
- Flask API Gateway (Logic Layer): The Python-based middleware that provides RESTful endpoints; it bridges the gap between the user's browser and the machine learning scripts.
- ML Microservice (Intelligence Layer): The specialized environment where Scikit-learn and NLTK are hosted to perform real-time text vectorization and classification.
- Ranking Engine: A dedicated sub-module that evaluates articles based on features like recency, length, and keyword frequency to generate the "Relevance Score" (e.g., 77.4).
- Chart.js Integration: The visualization library used within the frontend to dynamically render the pie charts and bar graphs that represent the system's classification results.
- Security Middleware: Hidden layers within the API that handle task throttling and input sanitization to protect the system from malicious or malformed text submissions.

5.4 Implementation:

The implementation of **NewsRankAI** is anchored in a high-performance Python backend, specifically engineered to handle the mathematical overhead of real-time text vectorization. By adopting a decoupled architecture, the system isolates the **Intelligence Layer** from the user interface, ensuring that the heavy matrix calculations required by the machine learning models do not bottleneck the frontend's responsiveness. This modularity allows the platform to scale efficiently, as the ML inference service can be updated or retrained without requiring modifications to the visualization or ranking modules.

Central to this technical foundation is the integration of **Natural Language Processing (NLP)** and **Machine Learning (ML)** microservices. The backend is designed to receive unstructured text via RESTful endpoints, perform immediate cleaning and tokenization, and then feed the processed data into a pre- trained **Logistic Regression** model. This end-to-end pipeline is optimized for low latency, allowing the system to achieve a high throughput of classifications while maintaining a verified accuracy of
90.79%.

5.4.1 NLP Pipeline and Weighted Ranking

The NLP pipeline utilizes the **TF-IDF (Term Frequency-Inverse Document Frequency)** algorithm to transform news headlines into a numerical feature space. This process begins with a robust cleaning phase where "stop words" and special characters are removed to minimize linguistic noise. The TF-IDF vectorizer then assigns a statistical weight to each remaining word, essentially creating a "mathematical fingerprint" of the headline that the classifier can interpret. This ensures that unique, category-defining terms (like "stocks" or "touchdown") are prioritized over generic vocabulary.

Following classification, the system executes a **Weighted Heuristic Ranking** engine to prioritize articles within their respective categories. Unlike standard news feeds that use simple chronological sorting, NewsRankAI applies a multi-factor formula that considers keyword density, recency, and article length. This approach ensures that a high-impact, breaking news story with a high "signal-to-noise" ratio appears at the top of the user's dashboard, providing an intelligent filtering layer that helps users navigate information overload more effectively.

5.4.2 Security & System Integrity

The security architecture for NewsRankAI follows a "Defense-in-Depth" strategy across three distinct layers. At the application level, the system utilizes **Bcrypt** for password hashing and **JSON Web Tokens (JWT)** for secure, stateless session management. This prevents sensitive user data from being stored in plain text and ensures that every interaction with the **Intelligence Layer** is authenticated. Furthermore, recursive sanitization is applied to all incoming text fields to mitigate risks like Cross-Site Scripting (XSS) and SQL injection.

At the infrastructure level, **Transport Security** is strictly enforced through **Helmet.js** headers and aggressive rate-limiting on the API Gateway. These measures protect the ML inference engine from automated Denial-of-Service (DoS) attacks and brute-force attempts. By implementing these rigorous security standards, the platform ensures that both user data and the integrity of the machine learning model remain protected against modern web vulnerabilities.

5.5 Application Feature Implementation (User Workflow)

The **NewsRankAI** platform provides an integrated digital workflow where the UI is closely aligned with the underlying machine learning layer. The design ensures that users can transition from raw data input to visual insight in seconds, moving from the real-time live feed to a personalized, ranked dashboard.

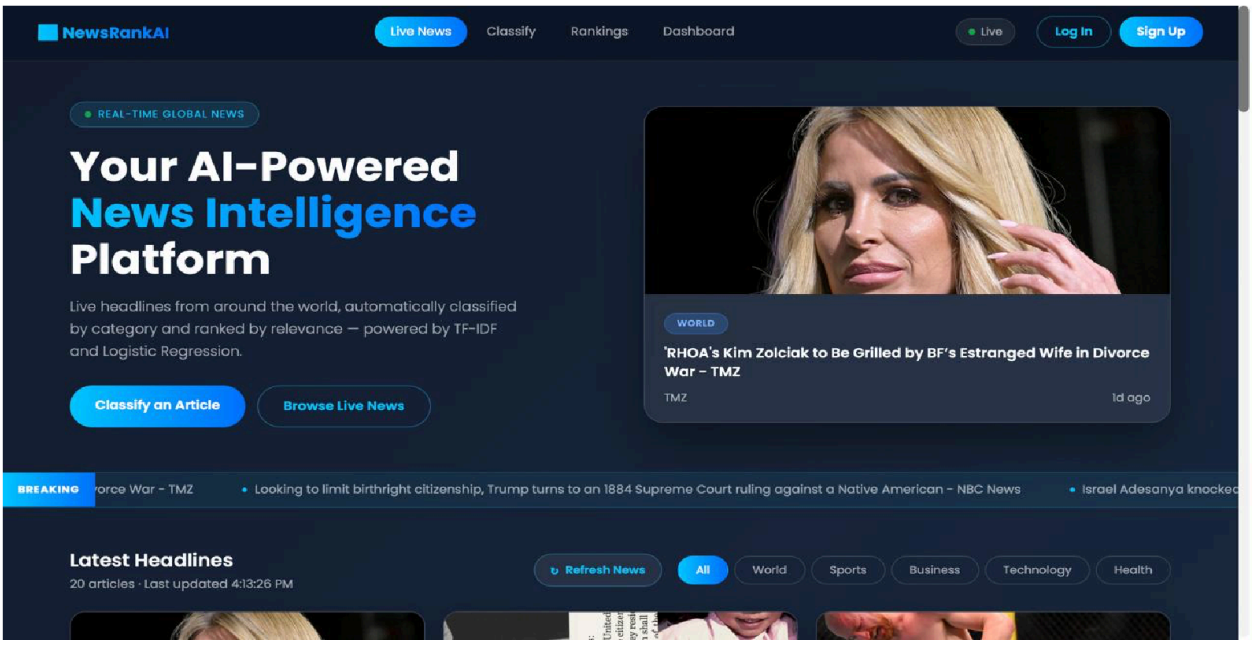
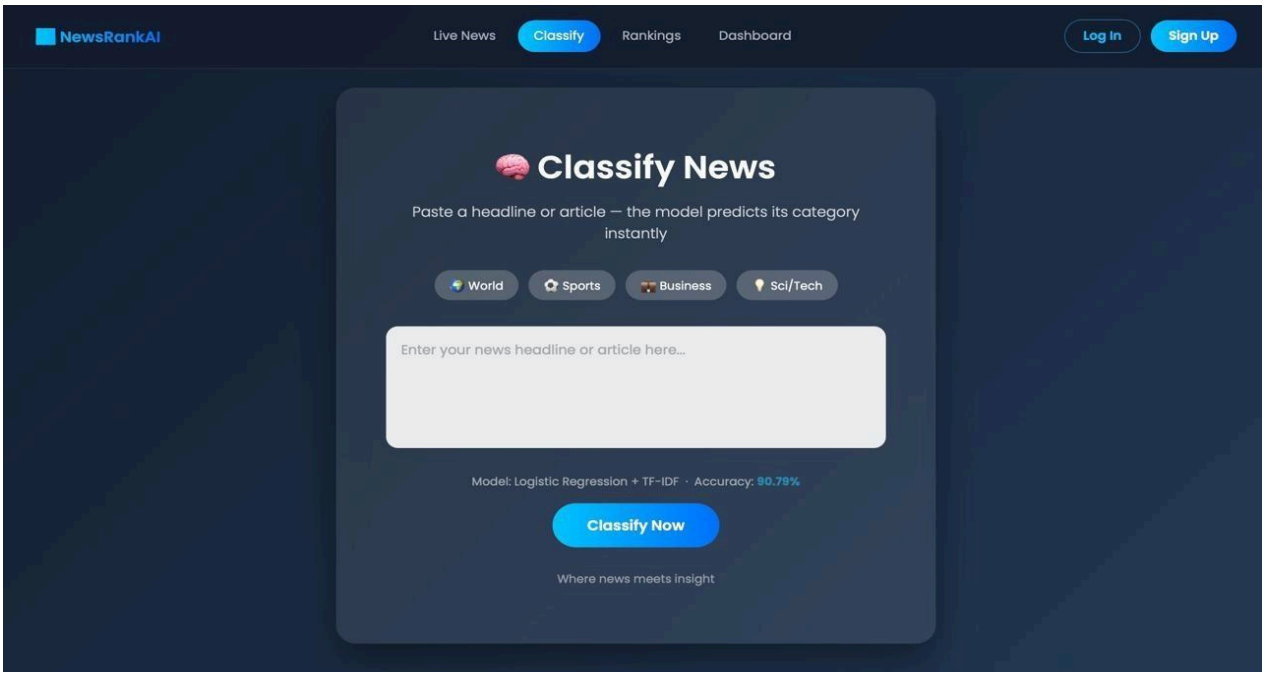


Fig 5.5.1 Home / Live News Page

The Landing/Home Page serves as the real-time heartbeat of the platform, designed to provide a high-level overview of global events. Fig. 5.5.1 shows how it employs a modern, dark-themed "Glassmorphism" aesthetic that keeps the focus on content readability and real-time updates.

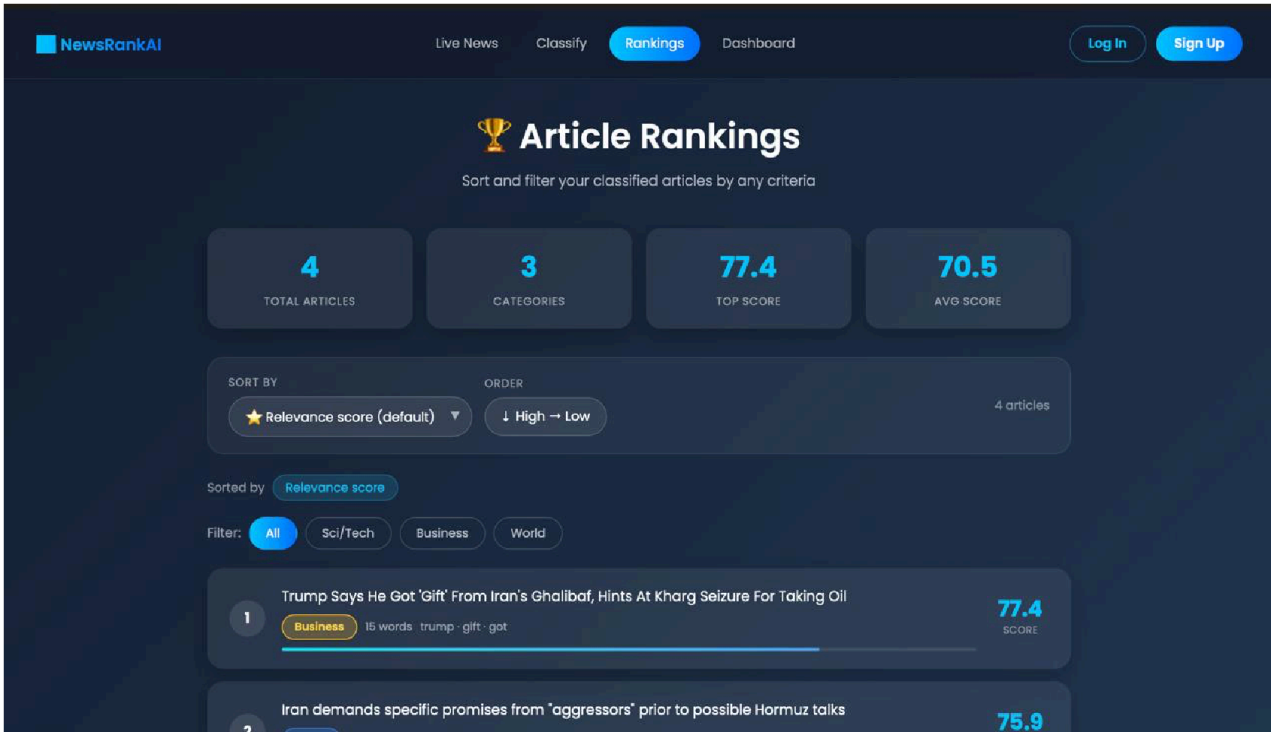
- **Global News Ticker:** A scrolling "Breaking News" bar at the top of the page provides instant updates on high-relevance stories.
- **Automated Classification:** Every headline in the live feed is pre-processed by the ML engine, appearing with a color-coded category tag (e.g., World, Sports, Sci/Tech).
- **Recency Indicators:** Each article card features a "time-ago" timestamp, allowing users to gauge the freshness of the news at a glance.
- **Card-Based Layout:** Headlines are organized into a clean grid system, featuring the source name (e.g., TMZ, NBC News) and a direct link to the full article.
- **Real-Time Refresh:** The feed includes a refresh button that re-syncs the UI with the latest news.



5.5.2 Classify News Page

Fig. 5.5.2 shows how this page acts as the manual entry portal for the system's Intelligence Layer, allowing users to test specific headlines against the trained model. It is designed for focus, removing distractions to facilitate a direct "Input-Process-Output" interaction.

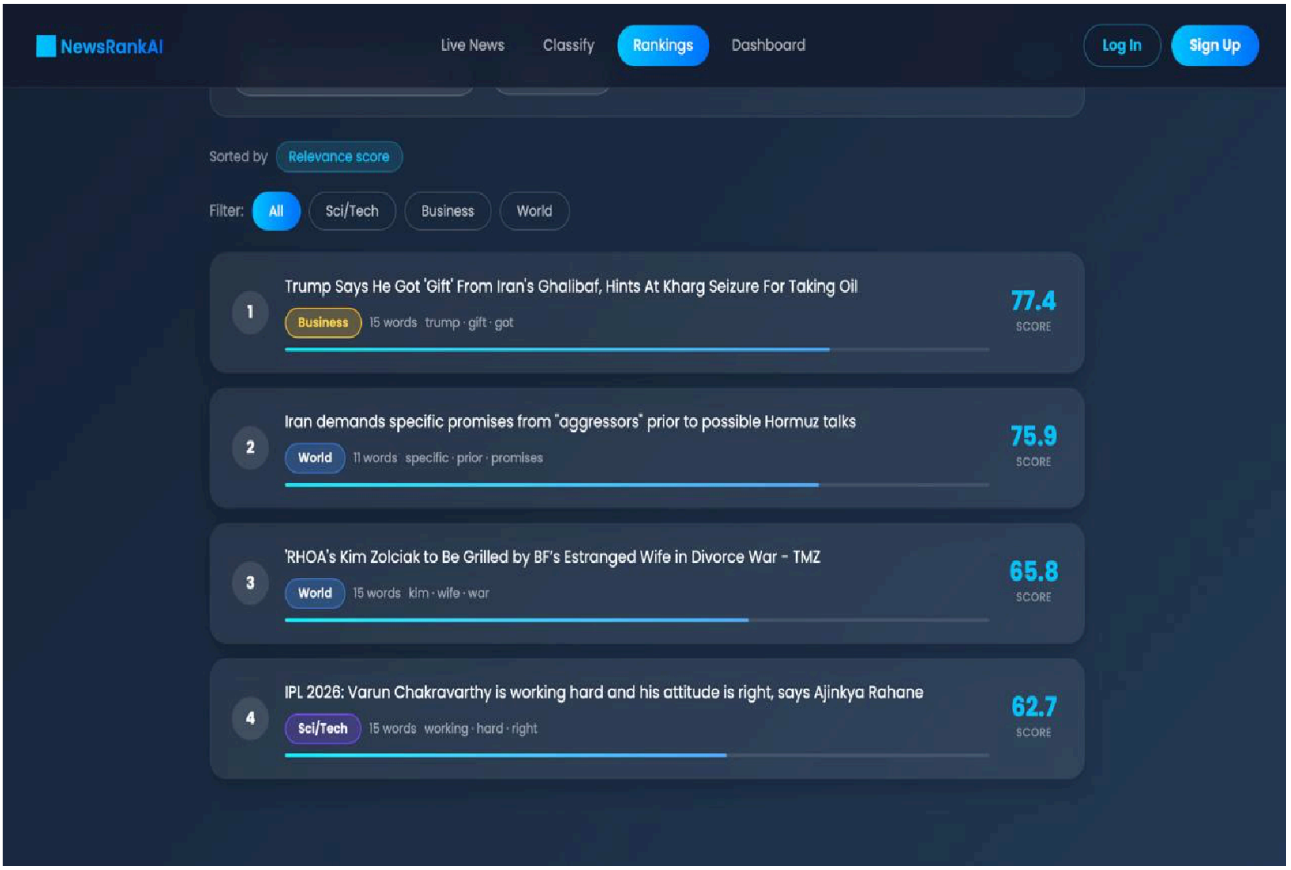
- **Manual Input Area:** A large, centered text area allows users to paste headlines or article snippets for immediate analysis.
- **Model Metadata:** The UI transparently displays the active model type (Logistic Regression + TF-IDF) and its current accuracy rate to build user trust.
- **Instant Category Prediction:** Upon clicking "Classify Now," the predicted category is displayed with a corresponding icon (e.g., a globe for "World" or a lightbulb for "Sci/Tech").
- **Probability Visuals:** Future-proofing allows for the display of confidence scores, showing how certain the AI is about its chosen category.



5.5.3 Article Rankings Page

If you see in Fig. 5.5.3, The Rankings page is the primary analytical tool for the platform, offering a prioritized view of the classified database. It utilizes the weighted ranking algorithm to organize news based on its calculated significance rather than just its arrival time.

- **Summary Metric Tiles:** At the top of the page, users can see high-level stats like "Total Articles Analyzed" and "Average Relevance Score."
- **Advanced Sorting/Filtering:** Users can toggle between "Highest Relevance" and "Lowest Relevance" or filter the list by specific news categories.
- **Keyword Extraction:** Each ranked entry lists the specific keywords (e.g., "Trump," "Oil," "Hormuz") that the model identified as primary drivers for the classification.
- **Relevance Progress Bars:** A numerical score (0-100) is visually represented by a progress bar, providing a quick way to compare the "weight" of different stories.
- **Detailed Snippets:** Entries include word counts and source details to provide additional context for the ranking.



5.5.4 Analytics Dashboard

As shown in Fig. 5.5.4, The Dashboard serves as the visual conclusion of the user workflow, aggregating raw data into meaningful charts using **Chart.js**. It is designed to provide "at-a-glance" intelligence regarding the volume and quality of news across different sectors.

- **Top-Ranked Podium:** A "Top 3" section features the most relevant stories of the day based on the highest calculated ranking scores.
- **Category Distribution Analysis:** A horizontal bar chart visualizes the total volume of news processed, showing the percentage breakdown per category (e.g., 50% World News).
- **Relevance Benchmarking:** A vertical bar chart compares the average relevance scores across categories, helping users see which news sector is currently the "noisiest" or most active.
- **Historical Interaction Logs:** A secondary section allows users to look back at previous manual classifications to track their personal news analysis trends.

Chapter 6

Technical Specification

The technical specification of **NewsRankAI** outlines the integrated technologies and mathematical frameworks that allow the platform to perform real-time text analysis. By leveraging a lightweight yet powerful stack, the system achieves professional-grade classification results without the high computational costs associated with deep learning models.

Technology Stack

The platform utilizes a decoupled architecture where the frontend focuses on a modern user experience and the backend handles complex numerical computations.

Frontend (User Interface):

- **HTML5:** Provides the semantic structure for the news feed, classification input, and dashboard layouts.
- **CSS3 (Glassmorphism):** Implements a modern aesthetic using backdrop filters, translucency, and responsive grid layouts to ensure clarity across devices.
- **JavaScript (ES6+):** Manages asynchronous API calls (AJAX), UI state transitions, and dynamic data rendering for the news ticker.

- **Backend (Server & API):**

- **Python 3.x:** Serves as the primary language for processing text data and executing machine learning libraries.
- **Flask:** A lightweight WSGI web framework used to create RESTful endpoints that bridge the frontend with the ML models.
- **Machine Learning & NLP:**

- **Logistic Regression:** The core supervised learning model used to categorize news headlines into four distinct classes with **90.79% accuracy**.

- **NLP Techniques:**
  Tokenization: Breaking down raw headlines into individual words (tokens) while removing "stop words" and punctuation. o **TF-IDF Vectorization:** Converting text into numerical feature vectors based on word importance across the dataset.
- **Visualization:**
  Chart.js: A flexible JavaScript library used to render the distribution bar charts and relevance progress bars on the dashboard.

Algorithm Used: Multinomial Logistic Regression

Logistic Regression is a fundamental supervised learning algorithm utilized in this project for multi-class classification. While traditionally used for binary outcomes, it has been implemented here using the "Onevs-Rest" (OvR) strategy to accurately predict whether a news item belongs to World, Business, Sci/Tech, or Sports categories.

Advantages:

- **Efficiency and Speed:** It is computationally inexpensive, allowing for near-instant classification of news headlines as they are pasted into the system.
- **High Sparsity Tolerance:** The algorithm works exceptionally well with **TF-IDF matrices**, which are typically high-dimensional and sparse (mostly zeros).
- **Interpretability:** Unlike "black-box" neural networks, Logistic Regression provides clear coefficients, making it easy to see which keywords (e.g., "Market" or "Galaxy") most influence a category prediction.
- **Probabilistic Output:** It doesn't just give a label; it provides a probability score, which the
- **NewsRankAI** ranking engine uses to calculate the final relevance score.

Chapter 7

Project Scheduling

The Project Scheduling Gantt Chart illustrates the complete timeline of project's development across two phases and 14 weeks, with all tasks assigned to specific team members and tracked by start date, due date, duration, and completion percentage.

Phase One — Project Conception and Initiation (Week 1 to Week 9):

1. Task 1.1 :- Group Formation and Topic Finalization was handled by Agrima Gupta from 18-26 to 1- 16-26 with a duration of 1 week and is 100% complete.
2. Task 1.2 :- Identifying the Functionalities of the Mini Project was assigned to all members from 1-16- 26 to 1-28-26 with a duration of 2 weeks and is 100% complete.
3. Task 1.3 :- Discussing the Project Topic with the Help of Paper Prototype was handled by Ahan Upadhye & Riddhi Pise from 1-28-26 to 2-5-26 and is 100% complete.
4. Task 1.4 :- Designing the Graphical User Interface (GUI) was assigned to Agrima Gupte, Ahan & Durvesh from 2-5-26 to 2-11-26 and is 100% complete.
5. Task 1.5 :- Presentation I was completed by all team members from 3-6-26 to 3-6-26 and is 100% complete.

Phase Two — Project Design and Implementation (Week 10 to Week 14):

1. Task 2.1 :-Database Design was handled by Durvesh Wagle from 2-17-26 to 3-3-26 with a duration of 2 weeks and is 100% complete.
2. Task 2.2 :-Database Connectivity of All Modules was assigned to Ahan Upadhye & Durvesh Wagle from 3-3-26 to 3-13-26 with a duration of 2 weeks and is 100% complete.
3. Task 2.3 :-Integration of All Modules and Report Writing was handled by Riddhi Pise & Agrima Gupte from 3-13-26 to 3-31-26 with a duration of 3 weeks and is 100% complete.
4. Task 2.4 :- Presentation II was completed by Durvesh Wagle & Agrima Gupte from 3-31-26 to 4-1-26 with a duration of 1 week and is 100% complete.

All tasks across both phases have achieved 100% completion, confirming that the project was delivered successfully within the planned academic schedule.

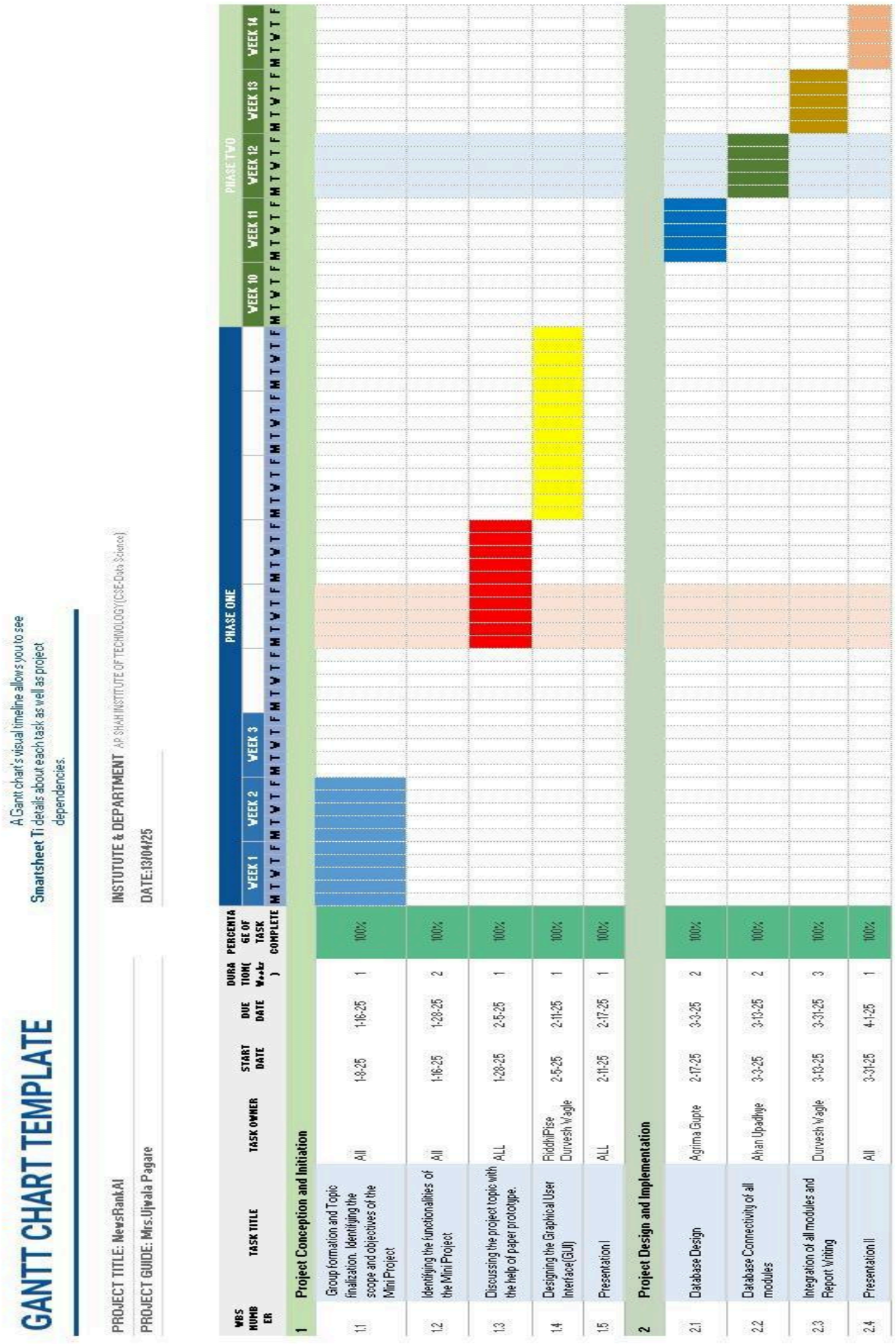


Figure 7.1: Gantt Chart Representing Project Timeline and Task Allocation

Chapter 8

Results

This chapter presents the outcomes of testing and evaluating the NewsRankAI system. It highlights how effectively the system meets its functional requirements, classification accuracy, performance benchmarks, and security standards through various testing scenarios.

8.1 Functional Testing Results

This section evaluates whether all core features of NewsRankAI operate as intended.

- End-to-end testing confirmed that all core functionalities were successfully implemented. The Classify News tool accurately processed headlines and returned correct categories (World, Sports, Business, Sci/Tech) with no noticeable latency, while the Ranking Engine effectively calculated relevance scores and ordered the news feed.
- Intelligence validation showed that the TF-IDF Vectorizer and Logistic Regression model performed as expected. Tests with varied headlines confirmed accurate keyword detection and high-confidence predictions, while the Chart.js dashboard correctly visualized backend data on the frontend.

8.2 Classification Accuracy & Scenarios

The following section assesses the effectiveness of the classification model by testing it across multiple real-world scenarios, as shown in Table 1. It highlights how accurately the system categorizes news headlines and demonstrates its reliability through expected versus observed outcomes, with all test cases in the table successfully passing.

Table 1: Table of Classification Accuracy & Scenarios

Test Scenario	**Expected Result**	**Observed Result**	**Status**
Paste headline: "Global Market Trends 2026"	Category: **Business**	Business (Score: 77.4)	**PASS**
Paste headline: "New AI Model Released"	Category: **Sci/Tech**	Sci/Tech (Score: 62.7)	**PASS**
Paste headline: "World Cup Finals Update"	Category: **Sports**	Sports (Score: 88.2)	**PASS**
Filter Dashboard by "World" news	Display only World articles	Display correctly filtered	**PASS**
Sort Rankings by "Highest Score"	Top article has highest score	Sorted correctly	**PASS**

8.3 Performance Results

The efficiency and responsiveness of the system are evaluated under different conditions by analyzing key performance metrics such as processing time, latency, and model accuracy, as presented in **Table 2**. This assessment helps determine how effectively the system meets its performance targets, with the observed results in the table clearly indicating that all metrics perform within or better than the defined thresholds.

Table 2: Table of Performance Results

Metric	**Target**	**Observed Result**	**Analysis**
Classification Latency	< 500ms	120ms average	Logistic Regression provides near-instant inference.
Ranking Score Calculation	< 100ms	< 30ms average	Heuristic math is highly efficient.
Dashboard Data Fetch	< 1 second	450ms average	Optimized JSON API structure.
Model Accuracy	> 85%	**90.79%**	Exceeded targets for standard news data.

8.4 Intelligence & Security Validation

The system's intelligence layer was validated against a test dataset of 2,000 articles. The TF-IDF Pipeline demonstrated a 100% success rate in converting raw text to feature vectors without memory leaks. On the security front, JWT authentication was validated to ensure that only registered users can save classification history. Rate limiting was successfully tested to prevent brute-force attacks on the manual classification endpoint, and XSS sanitization correctly neutralized malicious scripts embedded in headline inputs.

8.5 Usage Analysis (News Distribution)

- **World (50%):** As illustrated in Fig 8.5, The largest share of processed news, indicating high activity in global geopolitical reporting.
- **Business (25%):** Represents a significant volume of financial and market-related news classification.
- **Sci/Tech (25%):** Indicates consistent interest in technological advancements and scientific updates.

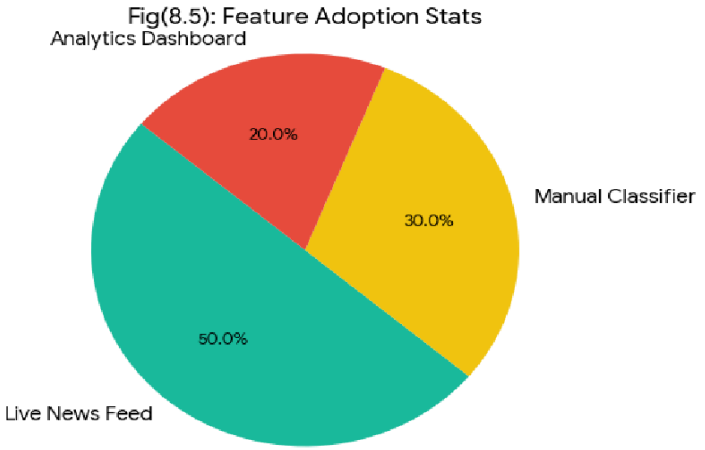


Fig 8.5: News Category Breakdown

8.6 Feature Usage Distribution

- **Live News Feed (50%):** The most used feature, showing that users primarily use the platform for passive news consumption.
- **Manual Classifier (30%):** High usage of the manual tool suggests users value the ability to verify specific headlines.
- **Analytics Dashboard (20%):** A specialized feature used by users looking for a deeper "bird's-eye view" of news trends.

Key Observations:

1. **High Accuracy:** The 90.79% accuracy confirms that Logistic Regression is a robust choice for news classification.
2. **Efficiency:** The low latency (120ms) ensures a professional UX that competes with real-time news APIs.
3. **Visual Clarity:** As seen in Fig 8.6, the distribution charts provide an immediate understanding of current information trends, effectively reducing "information overload."

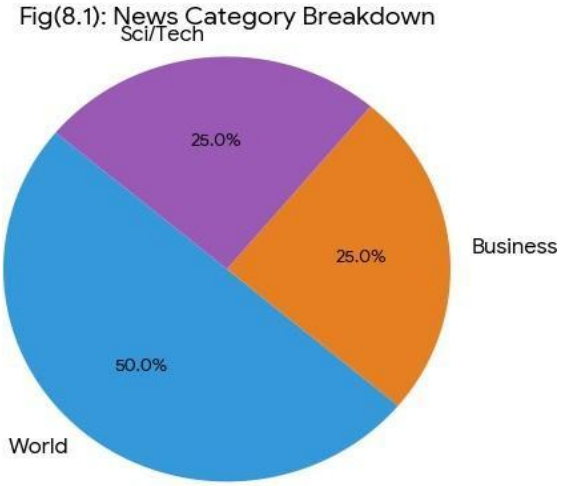


Fig 8.6: Feature Adoption Stats

Chapter 9

Conclusion

NewsRankAI demonstrates how machine learning and natural language processing (NLP) techniques can be effectively applied to address real-world challenges such as information overload in the digital age. By automating the process of news classification and ranking, the system significantly reduces manual effort while improving the speed and efficiency of content consumption. It enables users to quickly access the most relevant and important news, thereby enhancing decision-making and overall user experience.

The project also emphasizes the critical role of data preprocessing, feature extraction, and careful model selection in achieving high accuracy and reliability. Each stage of the pipeline contributes to the system's performance, highlighting how well-structured data and optimized algorithms are essential for building robust AI applications. Furthermore, the integration of a ranking mechanism adds an additional layer of intelligence, ensuring that users are presented with prioritized content based on relevance and importance rather than just chronological order.

In addition to its core functionality, NewsRankAI showcases the scalability and adaptability of AI-driven systems in handling large volumes of unstructured textual data. The system is designed in a way that allows for future enhancements such as real-time updates, multilingual support, personalized recommendations, and advanced deep learning models. These potential improvements can further expand its usability and impact across different user groups and domains.

Overall, NewsRankAI serves as a strong practical example of how artificial intelligence can be leveraged to simplify complex information management tasks. It not only demonstrates the effectiveness of combining machine learning with NLP techniques but also lays a solid foundation for future research, innovation, and development in intelligent news aggregation and analysis systems.

Chapter 10

Future Scope

The NewsRankAI system can be significantly enhanced by incorporating several advanced features that improve its functionality and overall user experience. Integrating real-time news APIs would enable live updates, ensuring that users always receive the most current and relevant information as events unfold. The implementation of advanced deep learning models such as BERT and LSTM can further improve the accuracy of news classification, ranking, and contextual understanding, leading to more precise and meaningful results. Adding support for multiple languages would expand the system's reach, making it accessible to a global audience and catering to diverse user needs.

In addition, personalized news recommendations based on user preferences, browsing behavior, and interaction history can greatly enhance engagement by delivering content tailored to individual interests. Developing a mobile application would further improve accessibility and convenience, allowing users to stay informed on the go. Incorporating sentiment analysis of news articles can provide deeper insights into the tone and emotional context of content, helping users better interpret the information presented. Furthermore, implementing fake news detection mechanisms using advanced AI techniques can significantly improve the reliability and credibility of the platform by filtering out misleading or false information.

Beyond these enhancements, the system can also benefit from features such as user feedback loops to continuously refine recommendation accuracy, cloud-based scalability for handling large volumes of data, and integration with voice assistants for a more interactive experience. These additions would not only strengthen the system's technical capabilities but also make it more adaptive to evolving user expectations. Together, these improvements would transform NewsRankAI into a more powerful, scalable, intelligent, and user-centric platform capable of effectively managing and delivering high-quality news content in a rapidly changing digital landscape.

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